

ENME601 Thermodynamics

Week 6 – The Second Law of Thermodynamics

Presenter: Dr. Jay Wang

Mid-semester Test time

On Friday, 21st August 2026 (in week 6, last day before mid-break)

Time: 6 pm – 9.15 pm (Please check your venues in your AUT Calendar)

This is a paper-based, **open-book** exam where **ALL** handwritten or printed notes are allowed. **Property Tables** will be provided.

Contents covered: Week 1 – Week 6.

Final version (6 questions = 30 min/question)

Aim: To assess your overall understanding of the content covered in Weeks 1–6.

Objectives

- Introduce the **second law of thermodynamics**.
- Discuss **thermal energy reservoirs, heat engines, refrigerators, and heat pumps** and their thermal efficiencies and coefficients of performance.
- Describe the **Kelvin–Planck** and **Clausius statements** of the second law of thermodynamics.
- Discuss **reversible and irreversible processes**.
- Describe the **Carnot cycle** and examine the **Carnot principles**, idealized Carnot heat engines, refrigerators, and heat pumps.
- Determine the expressions for the **thermal efficiencies** and **coefficients of performance** for **reversible heat engines, heat pumps, and refrigerators**.

Difference between "system efficiency" and "Coefficient of Performance" ?

Introduction

- The first Law requires that **energy be conserved during a process**. The Second Law of Thermodynamics goes further and asserts that **processes involving energy changes occur in a certain direction** and that **energy has quality as well as quantity**.
- A process cannot take place unless it **satisfies both the first law and the second law of thermodynamics**.
- These processes are okay according to the first law but violate the second law.

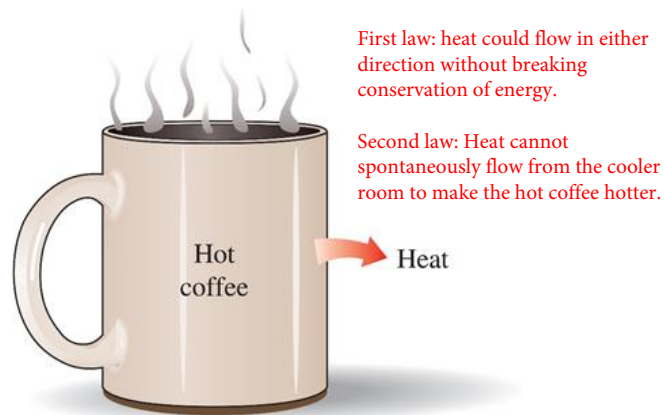


Figure 7-1: A cup of hot coffee does not get hotter in a cooler room.

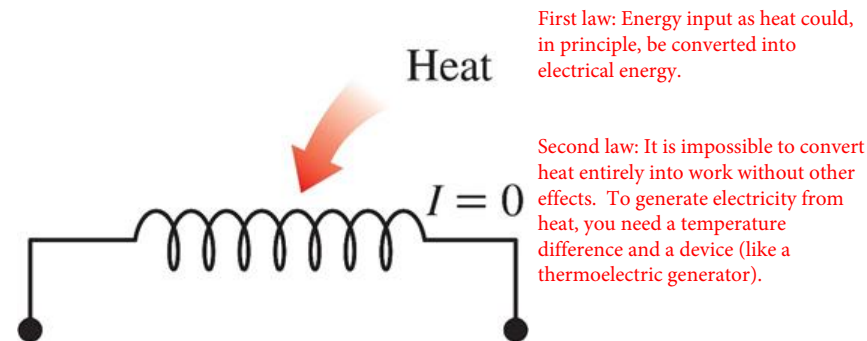


Figure 7-2: Transferring heat to a wire will not generate electricity.

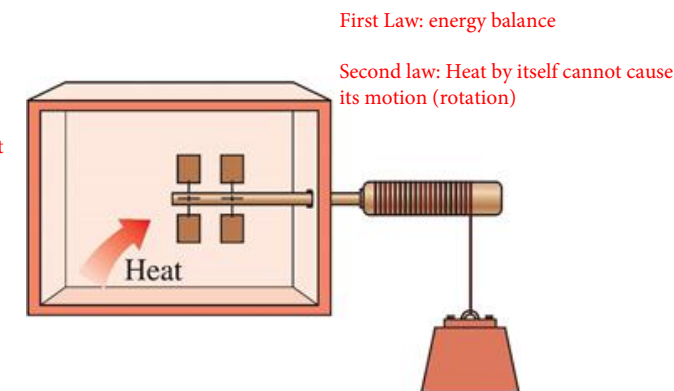


Figure 7-3: Transferring heat to a paddle wheel will not cause it to rotate.

Introduction

Not all heat can be converted into work. Some is always rejected to a cooler reservoir.

Heat $\rightarrow$ Work requires a device (like a heat engine) and a temperature difference.

Work $\rightarrow$ Heat, however, can happen fully (e.g., friction converts mechanical work entirely into heat).

Major uses of the second law.

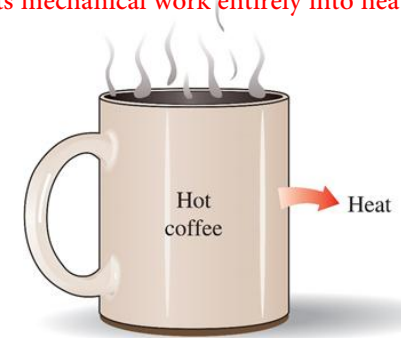
- The second law may be used to **identify the direction of processes**.
- The second law asserts **that energy has quality as well as quantity**. For example, we will find that work is a better form of energy flow than heat.
- The second law of thermodynamics is also used in determining the **theoretical limits for the performance** of commonly used engineering systems such as heat engines and refrigerators
- The second law can also predict the **degree of completion of chemical reactions**.

A V8 internal combustion engine.

Fuel Combustion (Heat Input): Burning gasoline releases chemical energy as high-temperature heat inside the cylinder.

Expansion (Work Output): The expanding gases push the piston down, doing mechanical work on the crankshaft.

Exhaust (Waste Heat Rejection): Not all heat can be converted into work, as it has exhaust gases.



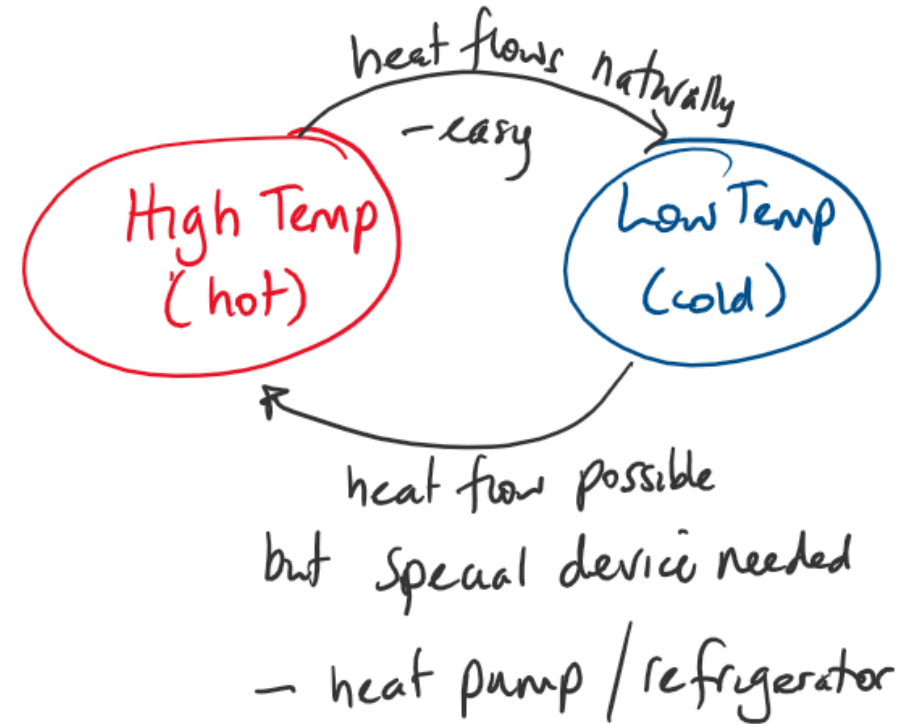
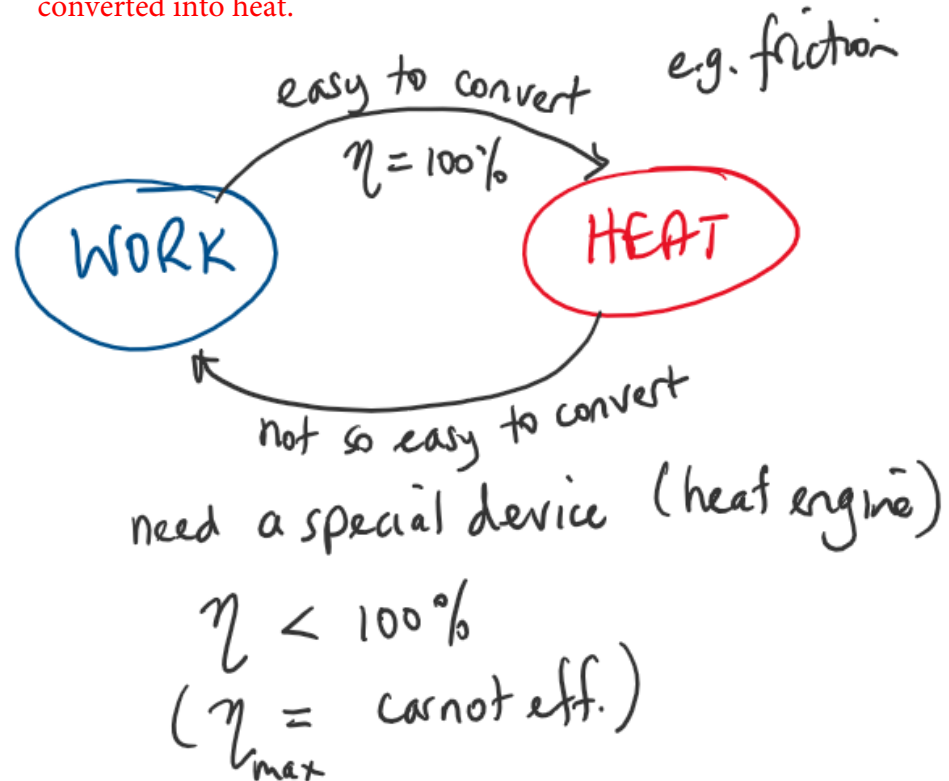
Heat from hot coffee to cold environment.



Introduction

Illustrations of the second law.

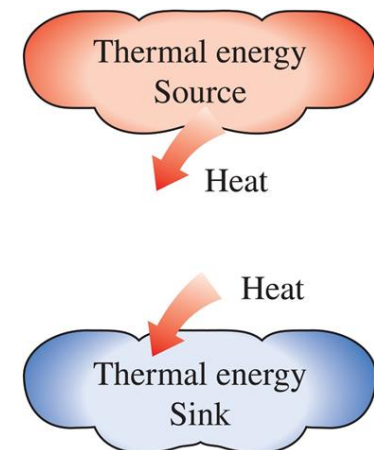
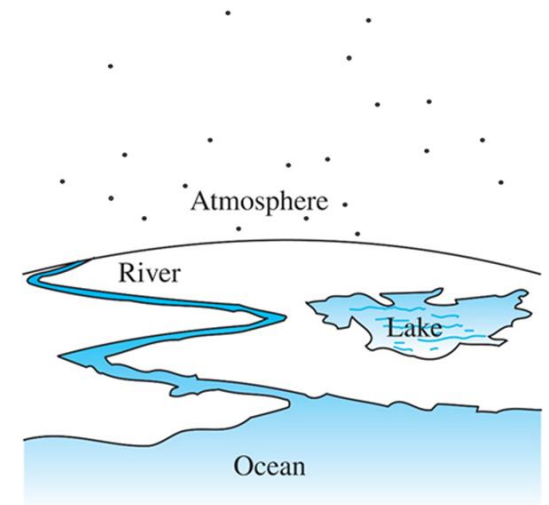
When friction occurs, mechanical energy (work) is completely converted into heat.



Carnot efficiency is a concept from thermodynamics that describes the **maximum possible efficiency** any heat engine can achieve when operating between two temperature reservoirs.

Thermal Energy Reservoirs (Sources and Sinks)

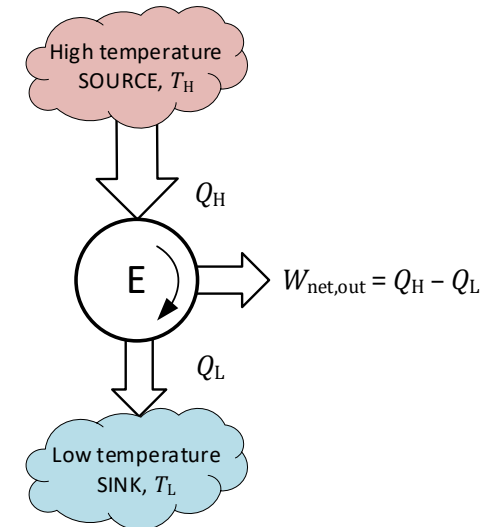
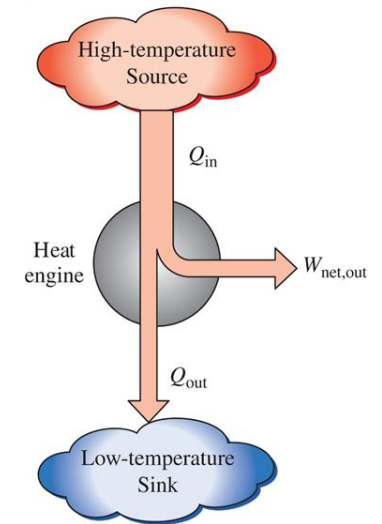
- A **thermal energy reservoir** (or just **reservoir**) is a hypothetical body that has a **large thermal energy capacity** (mass $\times$ specific heat)
- It can **supply or absorb finite amounts of heat** energy without undergoing any appreciable temperature change.
- Examples: the atmosphere, lakes, rivers, the ocean
- A reservoir that **supplies heat energy** is called a **source**
- A reservoir that **absorbs heat energy** is called a **sink**



Heat Engines

- Work can be converted directly and completely to heat (friction) but **converting heat to work requires a special device called a *heat engine***.
- Heat engines are characterised by
 - They receive heat, Q_H from a high-temperature source (at T_H). (combustion of fuel, solar energy, nuclear reactor etc).
 - They convert part of this heat to work, $W_{\text{net,out}}$ (e.g. rotating shaft).
 - They reject remaining waste heat, Q_L to a low-temperature sink (at T_L). (Usually the atmosphere, lake etc).
 - They operate on a cycle.
- Heat engines and other cyclic devices usually involve a fluid called the ***working fluid***. Heat is transferred to and from the working fluid as it undergoes the cycle.
- Note: $Q_H = Q_{\text{in}}$, $Q_L = Q_{\text{out}}$ (and are all magnitudes).

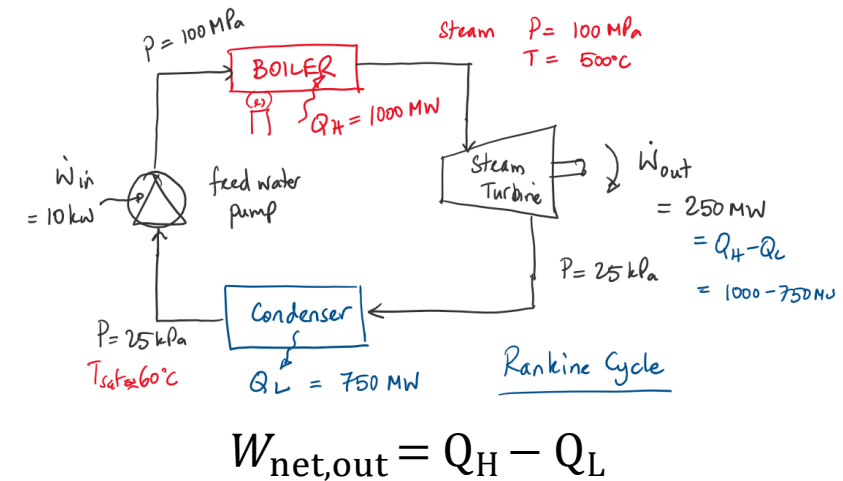
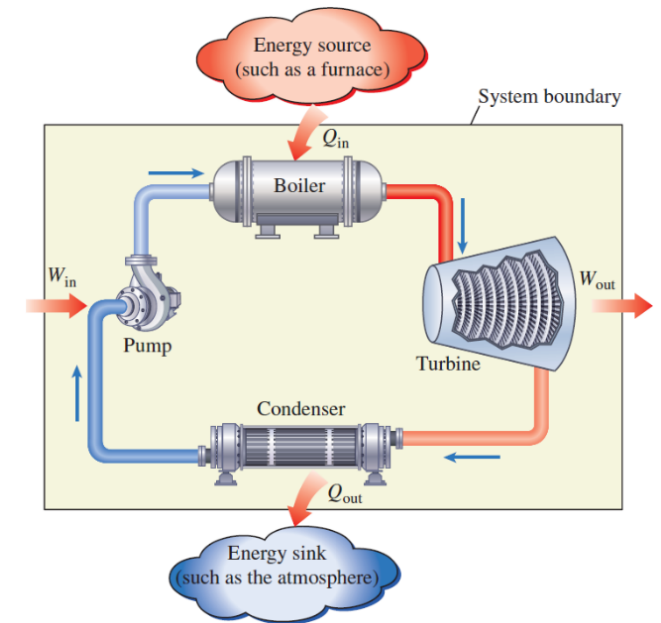
We are treating these heat transfers as positive quantities (absolute values).



Heat Engines

Example: Steam Power Plant (Rankine Cycle)

- Heat is supplied to the cycle ($\dot{Q}_H$) via the boiler (e.g. combustion of gas in the furnace). This increases the temperature and energy of the steam (e.g. 100 MPa, 500°C)
- The high energy steam enters the turbine which converts some of that energy into shaft work ($\dot{W}_{out}$) and leaves at low pressure
- The steam condenses to liquid water in the condenser by rejecting heat ($\dot{Q}_L$) to the environment.
- The liquid water at low pressure from the condenser enters the feed water pump which increases the pressure (compresses) the liquid water to boiler pressure ($\dot{W}_{in}$).



$$\dot{W}_{net,out} = \dot{Q}_H - \dot{Q}_L$$

The pump work (10 kW) is much smaller than the other quantities, so it can be neglected

Heat Engines

Thermal Efficiency

- A measure of the **performance** of a heat engine
- Generally we can define performance as:

$$\text{performance} = \frac{\text{desired output}}{\text{required input}}$$

- For a heat engine the desired output is $W_{\text{net,out}}$ and the required input is Q_H . So the thermal efficiency of a heat engine, η_{th} is:

$$\eta_{\text{th}} = \frac{W_{\text{net,out}}}{Q_H}$$

- And from the first law: $W_{\text{net,out}} = Q_H - Q_L$, so

$$\eta_{\text{th}} = \frac{Q_H - Q_L}{Q_H} = 1 - \frac{Q_L}{Q_H}$$

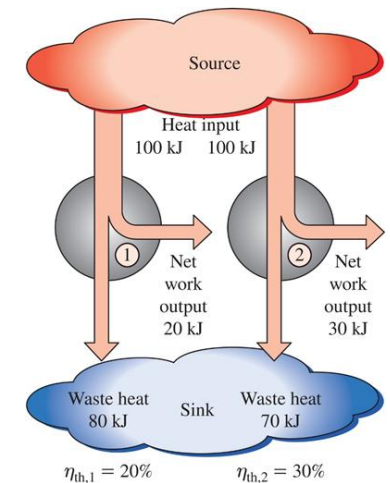
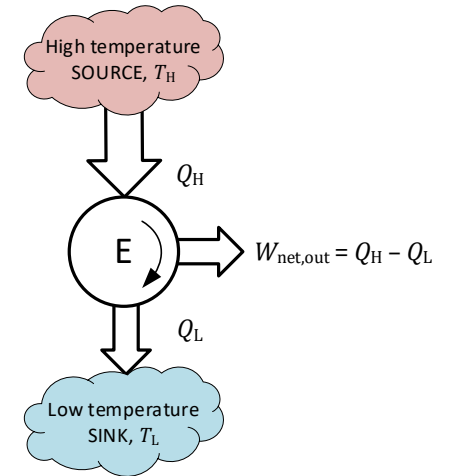


Figure 7-12: Some heat engines perform better than others (convert more of the heat they receive to work).

Heat Engines

Can we save Q_L (Q_{out})?

- For a heat engine to operate there **must** be some heat rejected (Q_L or Q_{out})
- Often that is a significant amount e.g. if thermal efficiency is 40% then 60% of the heat supplied is rejected as waste heat.
- That waste heat cannot be used usefully in the current cycle however it could be used in other processes. E.g.
 - for another heat engine where it becomes the heat source.
 - Process heat for other industrial operations.

Cascade Heat Engines

- The rejected heat (Q_L) from one cycle can serve as the heat input (Q_H) for another engine operating at lower temperatures.
- Example: A **combined cycle power plant** uses a gas turbine first (high T_H), then a steam Rankine cycle to use the hot exhaust gases.

Cogeneration (CHP – Combined Heat and Power)

- Instead of just rejecting heat to the environment, it can be supplied as **process heat** to factories, refineries, or chemical plants.
- Or it can be used for **district heating** (heating homes, offices, etc. in winter).

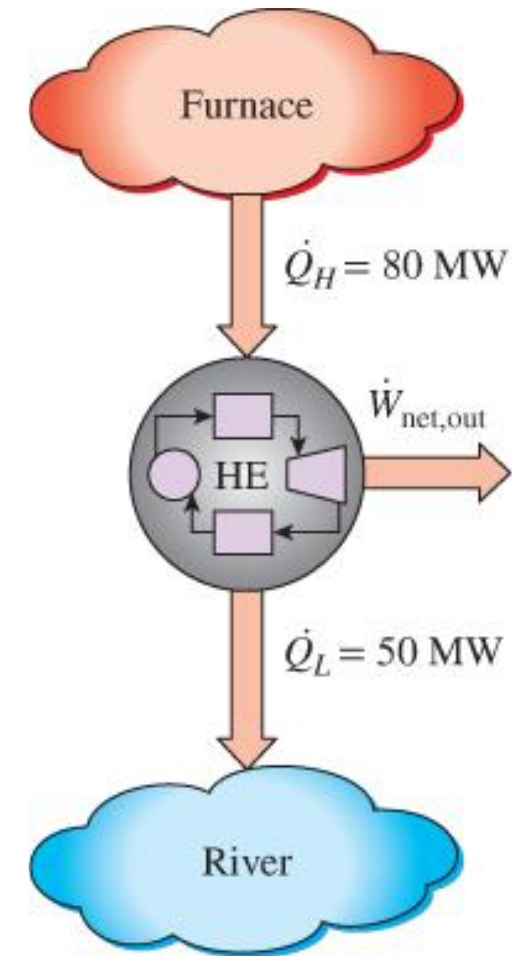
Absorption Refrigeration

- Waste heat can drive absorption chillers, which provide cooling without requiring electrical compressors.

Short practice 1:

Hints:

Heat is transferred to a heat engine from a furnace at a rate of 80 MW. If the rate of waste heat rejection to a nearby river is 50 MW, determine the net power output and the thermal efficiency for this heat engine.



- **Assumptions:** Heat losses through the pipes and other components are negligible.
- **Analysis:** A schematic of the heat engine is given in . The furnace shown in figure, serves as the high-temperature reservoir for this heat engine and the river as the low-temperature reservoir. The given quantities can be expressed as:

$$\dot{Q}_H = 80 \text{ MW} \quad \text{and} \quad \dot{Q}_L = 50 \text{ MW}$$

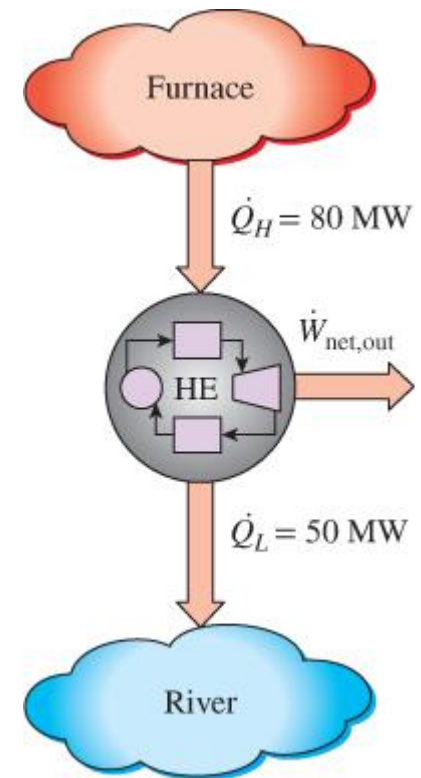
The net power output of this heat engine is

$$\dot{W}_{\text{net,out}} = \dot{Q}_H - \dot{Q}_L = (80 - 50) \text{ MW} = \mathbf{30 \text{ MW}}$$

Then the thermal efficiency is easily determined to be

$$\eta_{\text{th}} = \frac{\dot{W}_{\text{net,out}}}{\dot{Q}_H} = \frac{30 \text{ MW}}{80 \text{ MW}} = \mathbf{0.375} \text{ (or 37.5 percent)}$$

Discussion Note that the heat engine converts 37.5 percent of the heat it receives to work.

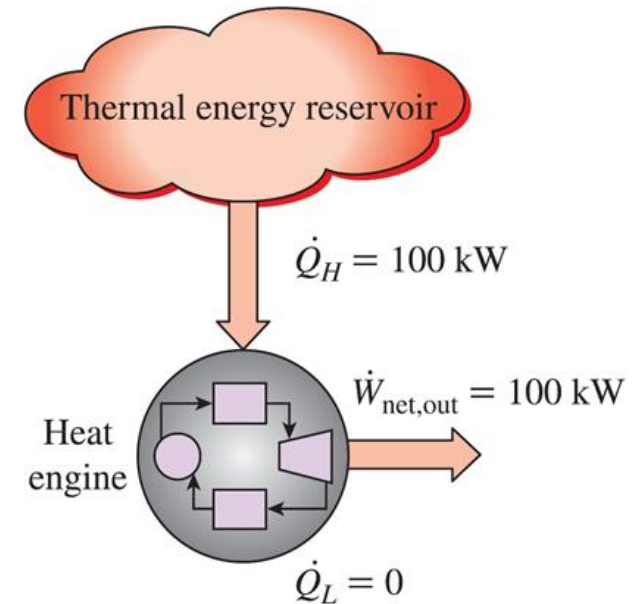


The Kelvin-Planck Statement

- **The Kelvin-Planck statement:**

It is impossible for any device that operates on a cycle to receive heat from a single reservoir and produce a net amount of work

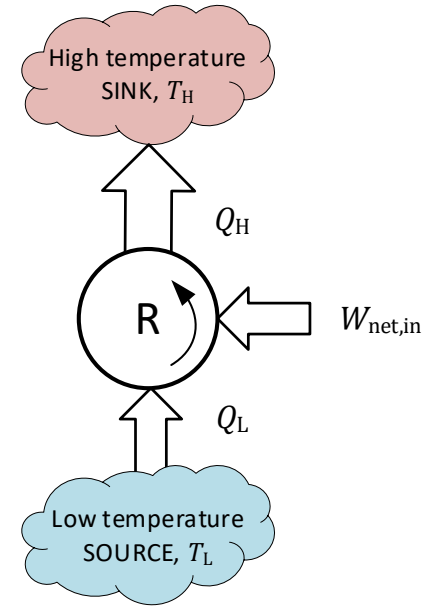
- In other words, it is impossible for Q_L to be zero and **impossible for the thermal efficiency to be 100%.**
- Or the thermal efficiency of a heat engine must be less than 100%.
- The Kelvin-Planck statement is a statement of the second law.



This is impossible.
(Okay by the first law)

Refrigerators and Heat Pumps

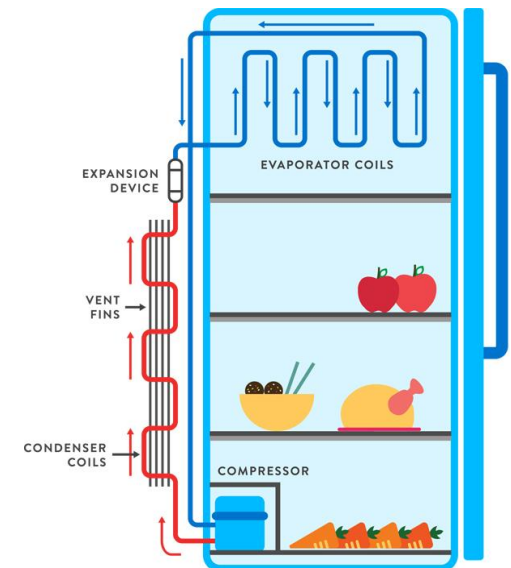
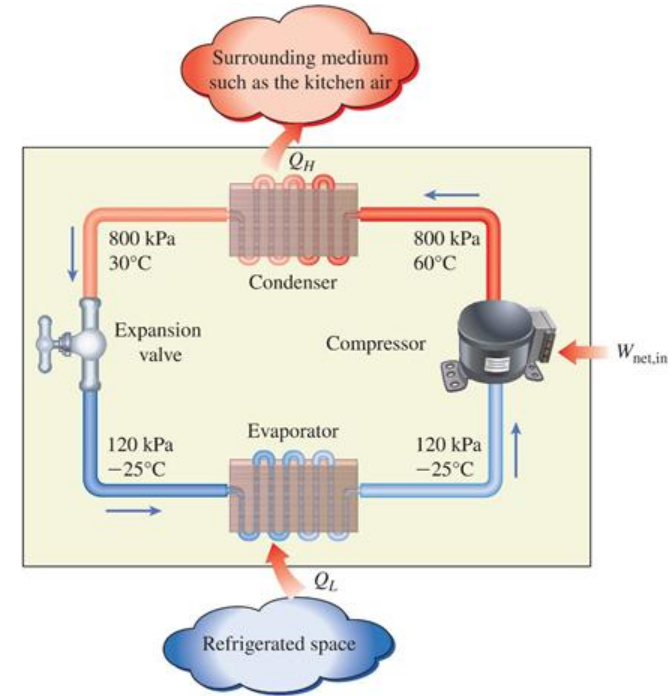
- Heat flows naturally from a high temperature (hot) to a low temperature (cold) but to cause **heat to flow from cold to hot** requires a special device, called a **refrigerator or heat pump**.
- The characteristics of a refrigerator or heat pump are:
 - Absorb heat (Q_L) from a low temperature source (at T_L).
 - Deliver heat (Q_H) to a high temperature sink (at T_H).
 - Requires energy input (usually work, $W_{\text{net,in}} = Q_H - Q_L$).
 - Operates in a cycle.
- The working fluid in these cycles is called a **refrigerant**



Refrigerators and Heat Pumps

Example: Vapour Compression refrigeration cycle

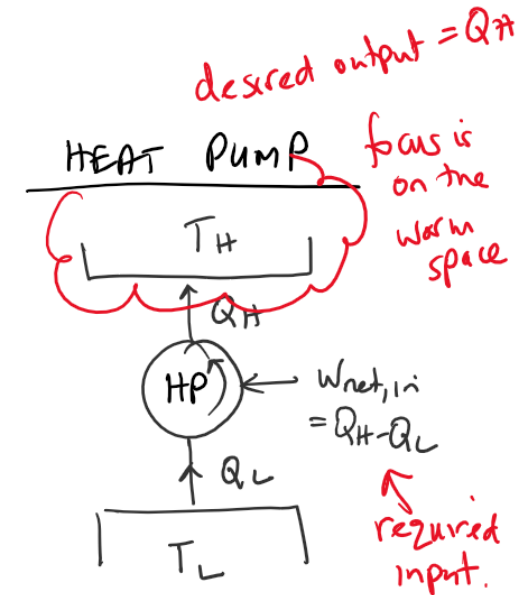
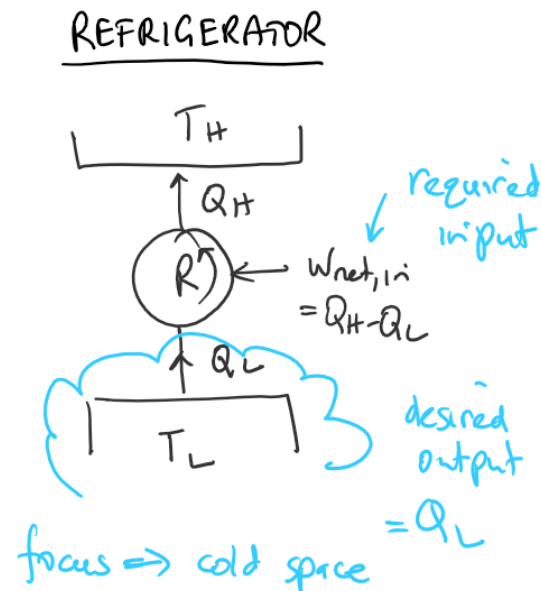
- Refrigerant enters the compressor as a vapour and is compressed to a higher pressure ($\dot{W}_{\text{net,in}}$).
- It leaves the compressor at relatively high temperature and condenses in the coils of the condenser by rejecting heat to the surroundings ($\dot{Q}_H$).
- It then enters a capillary tube where, as a result of the throttling effect, its pressure and temperature drop dramatically.
- The low temperature liquid refrigerant enters the evaporator where it evaporates by absorbing heat from the refrigerated space ($\dot{Q}_L$).
- The cycle is completed as the refrigerant leaves the evaporator as a vapour and re-enters the compressor.



Refrigerators and Heat Pumps

Heat Pump vs Refrigerator, what's the difference?

- They both operate on the same cycle but the difference is in terms of **purpose**:
 - Refrigerator: **focus is on the cold space**: keeps a cool space cold, so desired output is Q_L .
 - Heat pump: **focus is on the warm space**: keeps a warm space hot, so desired output is Q_H .



Refrigerators and Heat Pumps

The efficiency of a refrigerator or heat pump is expressed in terms of a coefficient of performance

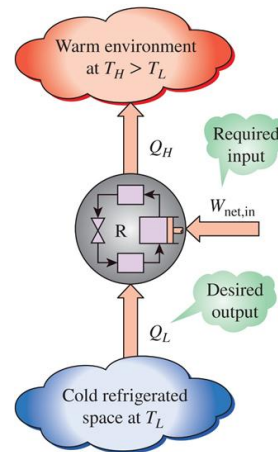
- **Refrigerator**: objective is to remove heat (Q_L) from the refrigerated space.

$$COP_R = \frac{\text{desired output}}{\text{required input}}$$

$$\Rightarrow COP_R = \frac{Q_L}{W_{\text{net,in}}}$$

And since $W_{\text{net,out}} = Q_H - Q_L$

$$COP_R = \frac{Q_L}{Q_H - Q_L} = \frac{1}{\frac{Q_H}{Q_L} - 1}$$



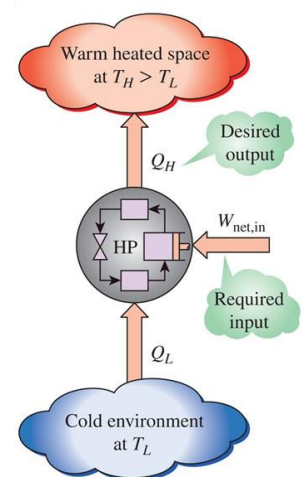
- **Heat pump**: objective is to deliver heat (Q_H) to a warmer space.

$$COP_{HP} = \frac{\text{desired output}}{\text{required input}}$$

$$\Rightarrow COP_{HP} = \frac{Q_H}{W_{\text{net,in}}}$$

And since $W_{\text{net,out}} = Q_H - Q_L$

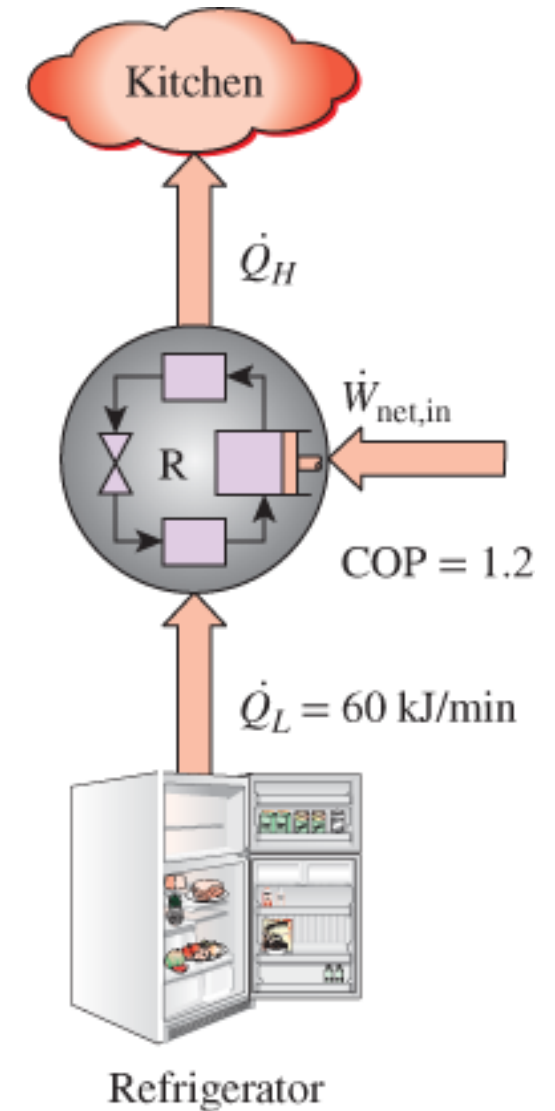
$$COP_{HP} = \frac{Q_H}{Q_H - Q_L} = \frac{1}{1 - \frac{Q_L}{Q_H}}$$



Short practice 2:

Hints:

A household refrigerator with a COP of 1.2 removes heat from the refrigerated space at a rate of 60 kJ/min. Determine (a) the electric power consumed by the refrigerator and (b) the rate of heat transfer to the kitchen air.

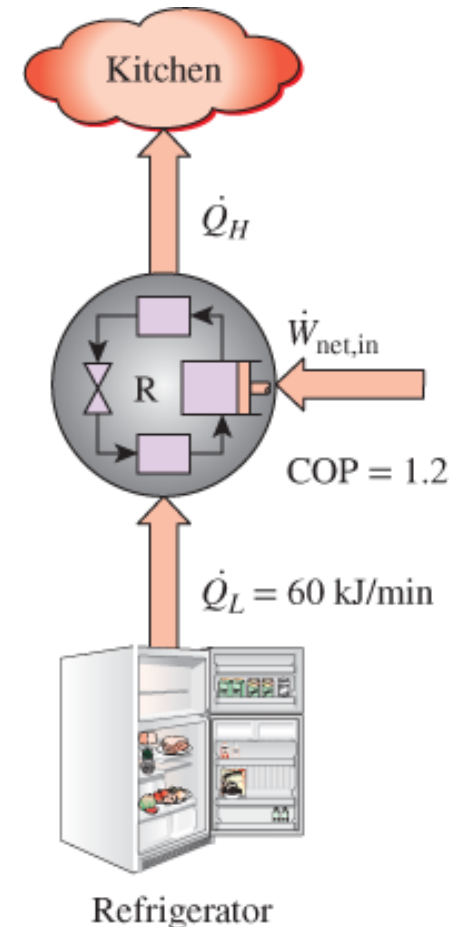


- **Assumptions** The refrigerator operates steadily.
- **Analysis** (a) Using the definition of the coefficient of performance, the power input to the refrigerator is determined to be

$$\dot{W}_{\text{net,in}} = \frac{\dot{Q}_L}{\text{COP}_R} = \frac{60 \text{ kJ/min}}{1.2} = 50 \text{ kJ/min} = \mathbf{0.833 \text{ kW}}$$

(b) The heat transfer rate to the kitchen air is determined from the energy balance,

$$\dot{Q}_H = \dot{Q}_L + \dot{W}_{\text{net,in}} = 60 + 50 = \mathbf{110 \text{ kJ/min}}$$



Short practice 3:

A heat pump is used to meet the heating requirements of a house and maintain it at 20°C . On a day when the outdoor air temperature drops to -2°C , the house is estimated to lose heat at a rate of $80,000 \text{ kJ/h}$. If the heat pump under these conditions has a COP of 2.5,

To determine:

- (a) the *a* power consumed by the heat pump and
- (b) the rate at which heat is absorbed from the cold outdoor air.

- **Assumptions** Steady operating conditions exist.
- **Analysis** (a) The power consumed by this heat pump is determined from the definition of the coefficient of performance to be

$$\dot{W}_{\text{net,in}} = \frac{\dot{Q}_H}{\text{COP}_{\text{HP}}} = \frac{80,000 \text{ kJ/h}}{2.5} = \mathbf{32,000 \text{ kJ/h}} \text{ (or 8.9 kW)}$$

(b) The house is losing heat at a rate of 80,000 kJ/h. If the house is to be maintained at a constant temperature of 20°C, the heat pump must deliver heat to the house at the same rate, that is, at a rate of 80,000 kJ/h. Then the rate of heat transfer from the outdoor becomes

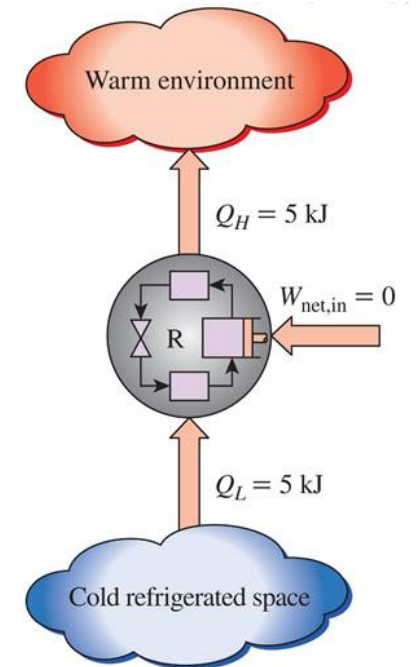
$$\dot{Q}_L = \dot{Q}_H - \dot{W}_{\text{net,in}} = (80,000 - 32,000) \text{ kJ/h} = \mathbf{48,000 \text{ kJ/h}}$$

The Clausius statement

The Clausius Statement:

It is impossible to construct a device that operates in a cycle and produces no effect other than transfer of heat from a lower temperature body to a higher temperature body

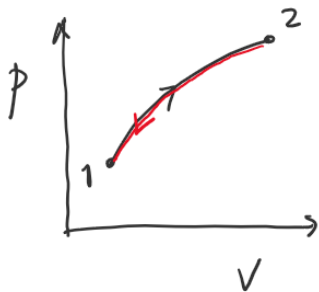
- In other words, it is impossible for a refrigerator or heat pump to operate unless there is some energy input (usually work) to drive the cycle
- It is impossible for the COP to be infinitely large (which it would be if $W_{\text{net,in}}$ was equal to zero: $COP_R = \frac{Q_L}{W_{\text{net,in}}}$).



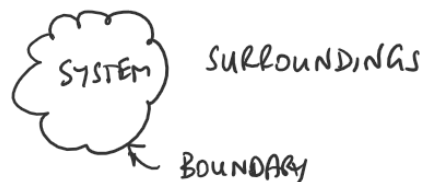
This is impossible.
(Okay by the first law)

Reversible and Irreversible Processes

- The second law states (Kelvin-Planck statement) that it is impossible for a heat engine to have a thermal efficiency of 100%. **So, what is the maximum possible efficiency of a heat engine?**
- To answer this question, we need to consider an **idealised process** called a **reversible process**.
- A reversible process is a process that can be **reversed with no net change to the system or surroundings**.
- **All normal, practical processes are irreversible** in that it may be possible to return the system to its original state but there would have been a net change on the surroundings.



1 → 2 a process
2 → 1 the reverse of that process



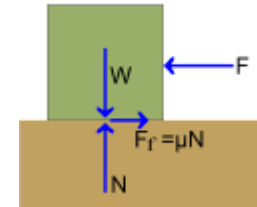
A process where, even if the system is restored to its initial state, the surroundings are left changed in some way.

Example: A car engine returns the piston to its starting position each cycle, but fuel is burned, heat is rejected to the atmosphere.

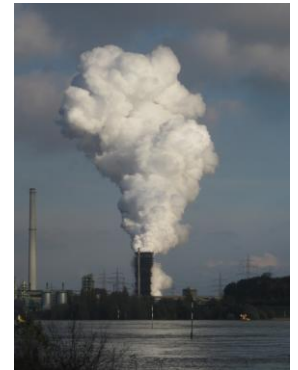
Reversible and Irreversible Processes

Irreversibilities

- These are factors that **cause ALL real processes to be irreversible** and they include:
 - friction
 - unrestrained expansion of gases
 - mixing of two fluids
 - heat transfer across a finite temperature difference
 - combustion, explosions
 - electric resistance
 - chemical reactions etc.
- In real, practical processes one or more of these factors will always be present.



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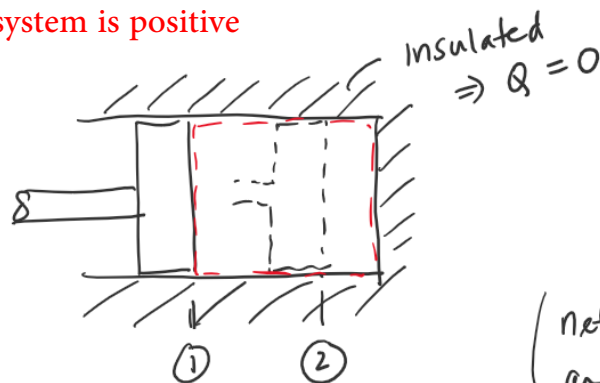
Reversible and Irreversible Processes

Reversible Processes

- The adiabatic compression of a gas in a cylinder comes close to being a reversible process.
- If we assume perfect insulation and frictionless operation, then it is a reversible process.
- In the practical process, friction (and heat loss) leave a net change on the surroundings and so is irreversible.

work done on the system is negative

work done by the system is positive



ideal (work)	practical (work)
① → ② - 8 kJ	① → ② - 9 kJ
② → ① + 8 kJ	② → ① + 7 kJ
net work 0 kJ	net change - 2 kJ
(net change on system and surroundings = '0')	(on system and surroundings)

REVERSIBLE
PROCESS

IRREVERSIBLE
PROCESS

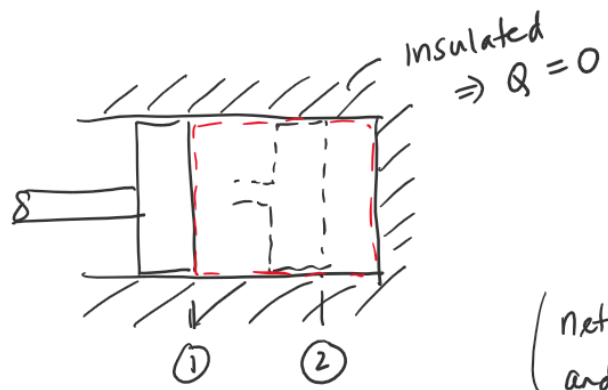
(need more work to overcome piston friction). $-8-1=-9$ kJ

(get less work back because friction resists motion) $8-1=7$ kJ

The system itself returns to its starting state, but the surroundings have lost 2 kJ as wasted work (dissipated as heat in friction).

Reversible and Irreversible Processes

- From this example it can be seen that reversible processes:
 - Consume the least amount of work (when the work is in) $8 \text{ kJ} < 9 \text{ kJ}$
 - Produce the maximum amount of work (when work is out) $8 \text{ kJ} > 7 \text{ kJ}$
- Therefore, a **heat engine cycle based on reversible processes should give us the maximum efficiency possible** (i.e. the theoretical limit for the thermal efficiency of a heat engine).

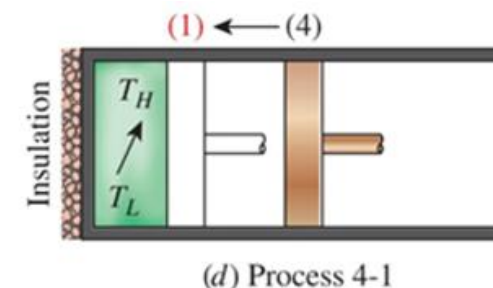
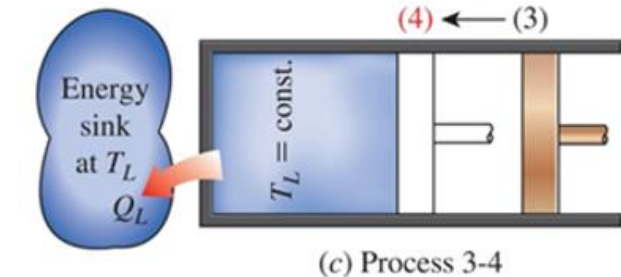
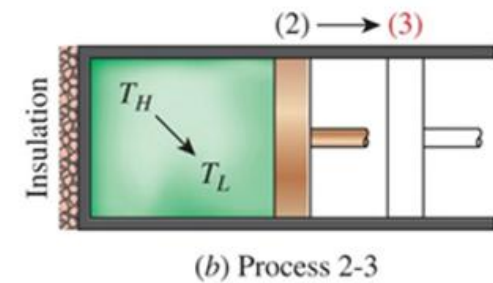
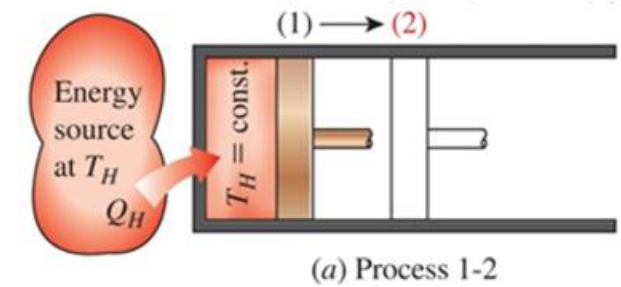


ideal (work)	practical (work)
$① \rightarrow ②$ -8 kJ	$① \rightarrow ②$ -9 kJ
$② \rightarrow ①$ $+8 \text{ kJ}$	$② \rightarrow ①$ $+7 \text{ kJ}$
net work 0 kJ	net change -2 kJ
(net change on system and surroundings = '0')	(on system and surroundings)
REVERSIBLE PROCESS	IRREVERSIBLE PROCESS

friction = 1 kJ

The Carnot Cycle

- The Carnot Cycle is a cycle based on reversible processes and is therefore **the most efficient cycle for a heat engine**.
- It is made up of four internally reversible processes:
 - 1 → 2 **frictionless isothermal absorption of heat** energy at the higher temperature, T_H . Work done, expansion process.
 - 2 → 3 **frictionless adiabatic expansion** to the lower temperature, T_L . Work done
 - 3 → 4 **frictionless isothermal rejection of heat** energy at the lower temperature, T_L . Work in, compression process
 - 4 → 1 **frictionless adiabatic compression** to the higher temperature, T_H .



The Carnot Cycle

- The Carnot Cycle illustrated on a $P - V$ diagram

- $1 \rightarrow 2$ frictionless isothermal absorption of heat energy
- $2 \rightarrow 3$ frictionless adiabatic expansion.
- $3 \rightarrow 4$ frictionless isothermal rejection of heat energy
- $4 \rightarrow 1$ frictionless adiabatic compression

Q_H : Heat absorbed by the system from the high-temperature reservoir at T_H .

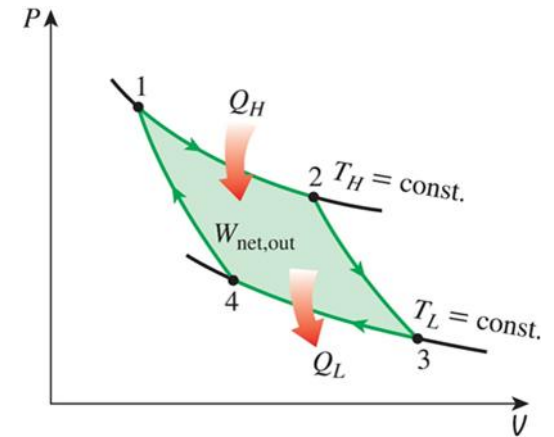
Q_L : Heat rejected by the system to the low-temperature reservoir at T_L .

- The Carnot cycle can be reversed (reversible processes!) in which case it becomes The Carnot refrigeration cycle

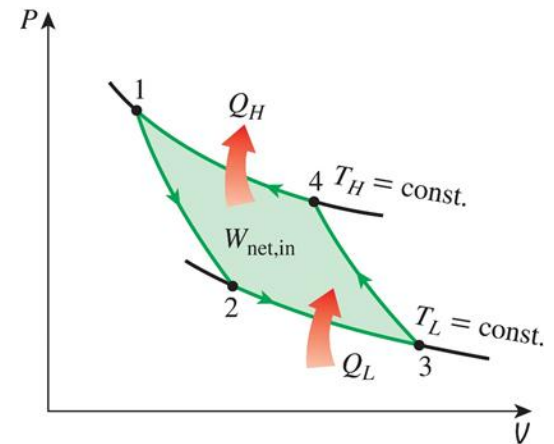
Reversed Carnot cycle:

Q_H : Heat rejected by the system to the high-temperature reservoir at T_H .

Q_L : Heat absorbed by the system from the low-temperature reservoir at T_L .



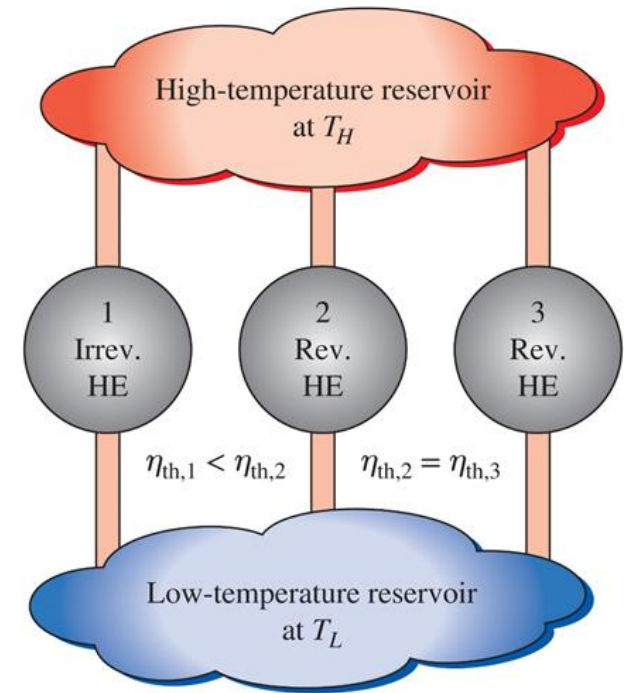
Carnot Cycle



Reversed Carnot Cycle

The Carnot Principles

- From the Kelvin-Planck and Clausius statements applied to reversible and irreversible heat engines it is possible to express the **Carnot Principles**:
 - The efficiency of an irreversible heat engine is always less than the efficiency of a reversible one operating between the same two reservoirs.
 - The efficiencies of all reversible heat engines operating between the same two reservoirs is the same.



Carnot Heat Engine – Carnot Efficiency

- It is possible to show that for a reversible cycle

$$\left(\frac{Q_H}{Q_L}\right)_{\text{rev}} = \frac{T_H}{T_L}$$

(Where temperatures are in Kelvin)

- Now, the efficiency of *any* heat engine is

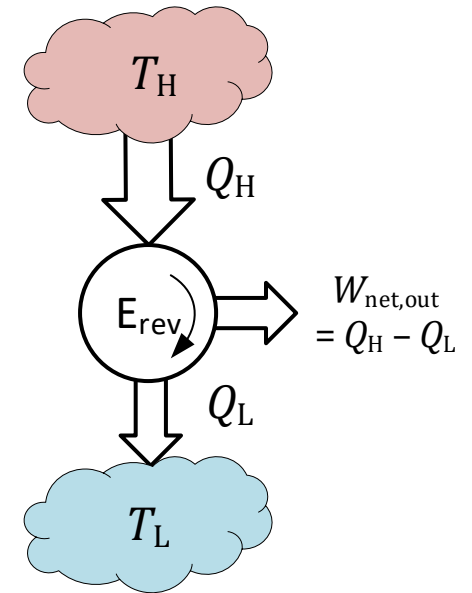
$$\eta_{\text{th}} = \frac{W_{\text{net,out}}}{Q_H} = \frac{Q_H - Q_L}{Q_H} = 1 - \frac{Q_L}{Q_H}$$

- So, the **efficiency of a Carnot or reversible heat engine is:**

$$\eta_{\text{th,rev}} = 1 - \left(\frac{Q_L}{Q_H}\right)_{\text{rev}}$$

So

$$\eta_{\text{th,rev}} = 1 - \frac{T_L}{T_H}$$



Carnot Heat Engine – Carnot Efficiency

$$\eta_{\text{th,rev}} = 1 - \frac{T_L}{T_H}$$

- This is the highest possible efficiency of a heat engine operating between two thermal energy reservoirs at T_H and T_L .
- And so, to maximize this ideal efficiency, the heat should be absorbed at as high a temperature as possible and rejected at as low a temperature as possible. This principle applies to practical, irreversible heat engines as well.
- So, if η_{th} is the thermal efficiency of a heat engine then:

If an engine is more efficient than the Carnot engine. That would violate the second law of thermodynamics, so it is impossible.

$$\eta_{\text{th}} \begin{cases} < \eta_{\text{th,rev}} & \text{irreversible (i.e. real) heat engine} \\ = \eta_{\text{th,rev}} & \text{reversible heat engine} \\ > \eta_{\text{th,rev}} & \text{impossible heat engine} \end{cases}$$

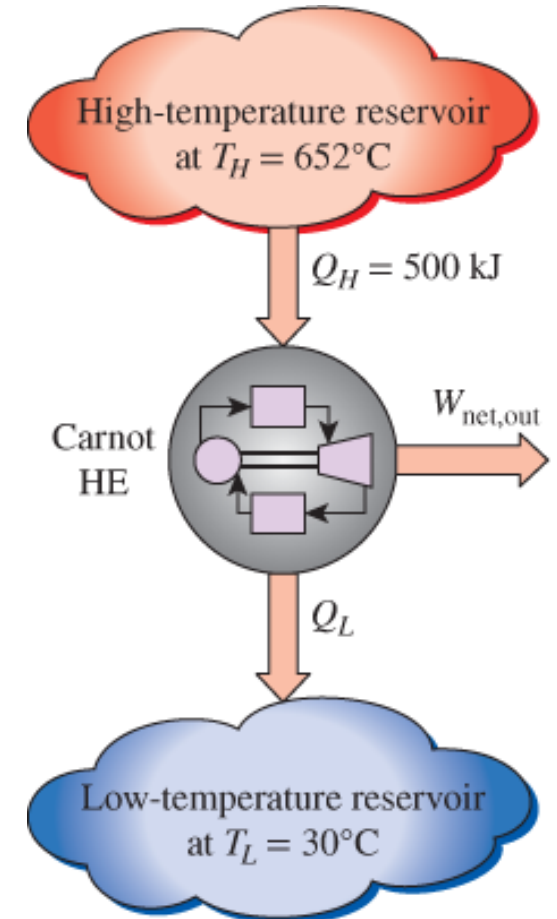
Any real (irreversible) engine has a lower efficiency than the reversible one operating between the same two reservoirs.

Short practice 4:

A Carnot heat engine receives 500 kJ of heat per cycle from a high-temperature source at 652°C and rejects heat to a low-temperature sink at 30°C .

To determine:

- (a) the thermal efficiency of this Carnot engine
- (b) the amount of heat rejected to the sink per cycle.



Solution:

Analysis (a) The Carnot heat engine is a reversible heat engine, and so its efficiency can be determined from Eq. 6–18 to be

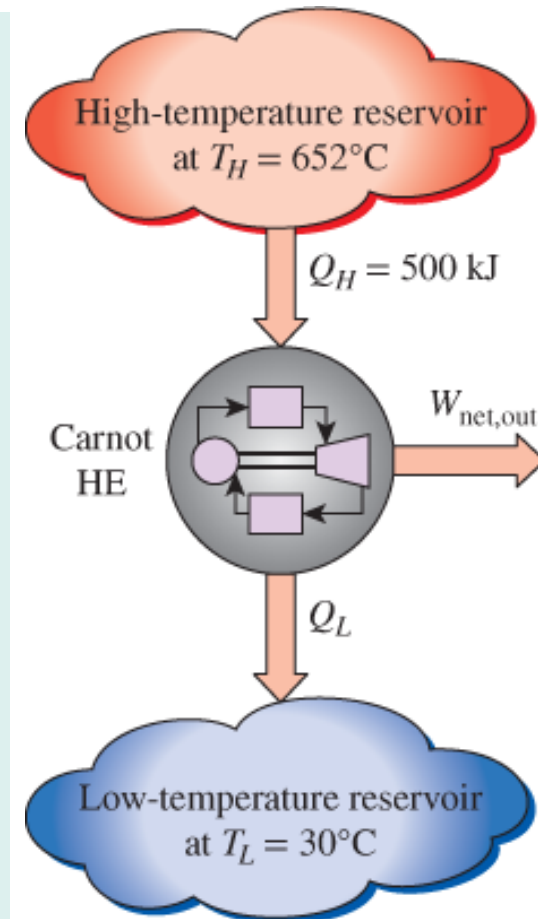
$$\eta_{\text{th,rev}} = 1 - \frac{T_L}{T_H} = 1 - \frac{(30 + 273) \text{ K}}{(652 + 273) \text{ K}} = \mathbf{0.672}$$

That is, this Carnot heat engine converts 67.2 percent of the heat it receives to work.

(b) The amount of heat rejected Q_L by this reversible heat engine is easily determined from Eq. 6–16 to be

$$Q_{L,\text{rev}} = \frac{T_L}{T_H} Q_{H,\text{rev}} = \frac{(30 + 273) \text{ K}}{(652 + 273) \text{ K}} (500 \text{ kJ}) = \mathbf{164 \text{ kJ}}$$

Discussion Note that this Carnot heat engine rejects to a low-temperature sink 164 kJ of the 500 kJ of heat it receives during each cycle.



Quality of Energy

- We see that for the Carnot efficiency, the higher the temperature of the source the greater the thermal efficiency so *as the temperature of the high-temperature source increases more of the high-temperature thermal energy can be converted to useful work.*
- Therefore: **the higher the temperature the higher the quality of the heat energy.**
- And again, **work is a more valuable form of energy transfer than heat** as 100% of work can be converted to heat but **only a fraction of the heat** can be converted to work.

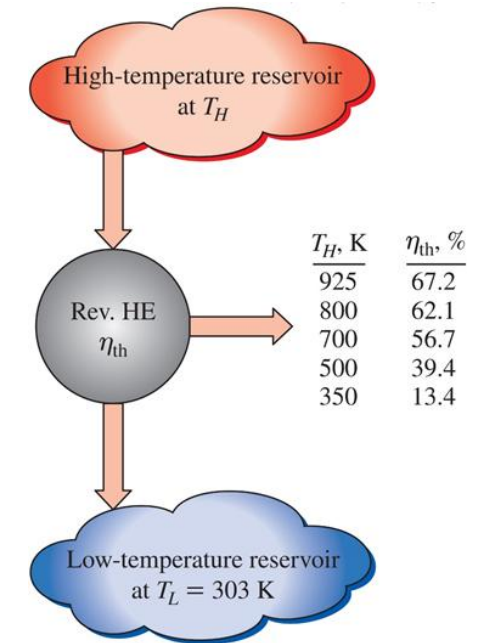
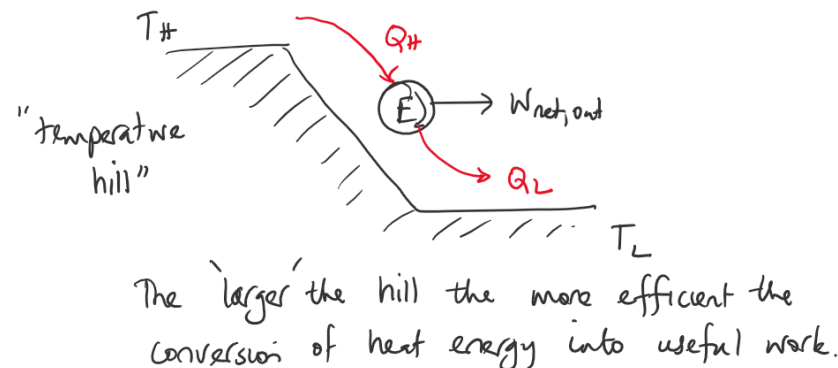


Figure 7–46: The fraction of heat that can be converted to work as a function of source temperature (for $T_L = 303 \text{ K}$).

Carnot Refrigerator and Heat Pump

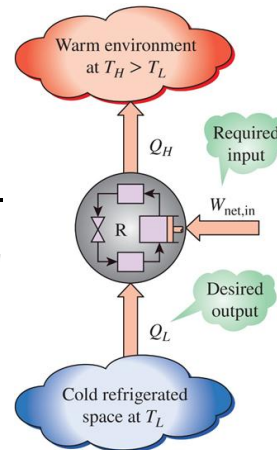
- If we consider a refrigerator or heat pump based on the reversed Carnot cycle, we can determine the **maximum possible COP** .

- Refrigerator.**

For any refrigerator:

$$COP_R = \frac{Q_L}{W_{\text{net,in}}} = \frac{Q_L}{Q_H - Q_L}$$

$$COP_R = \frac{1}{\frac{Q_H}{Q_L} - 1}$$

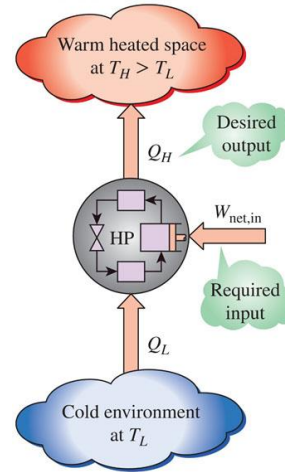


- Heat pump.**

For any heat pump:

$$COP_{HP} = \frac{Q_H}{W_{\text{net,in}}} = \frac{Q_H}{Q_H - Q_L}$$

$$COP_{HP} = \frac{1}{1 - \frac{Q_L}{Q_H}}$$



So for a **reversible refrigerator**

$$COP_{R,\text{rev}} = \frac{1}{\frac{T_H}{T_L} - 1}$$

So for a **reversible heat pump**

$$COP_{HP,\text{rev}} = \frac{1}{1 - \frac{T_L}{T_H}}$$

Carnot Refrigerator and Heat Pump

$$COP_{R,rev} = \frac{1}{\frac{T_H}{T_L} - 1}$$

$$COP_{HP,rev} = \frac{1}{1 - \frac{T_L}{T_H}}$$

- These COP 's represent the highest possible for a refrigerator or heat pump operating between the temperature limits of at T_L and T_H
- And so, to maximise these ideal coefficients of performance the difference between T_L and T_H should be as low as possible (a small “temperature hill”). In other words, the COP will reduce as the temperature difference increases. This applies to practical, irreversible heat engines as well.



Carnot Refrigerator and Heat Pump

- So, if COP_R is the coefficient of performance for a refrigerator then:

$$COP_R \begin{cases} < COP_{R,rev} & \text{irreversible (i.e. real) refrigerator} \\ = COP_{R,rev} & \text{reversible refrigerator} \\ > COP_{R,rev} & \text{impossible refrigerator} \end{cases}$$

- Similarly for a heat pump:

$$COP_{HP} \begin{cases} < COP_{HP,rev} & \text{irreversible (i.e. real) heat pump} \\ = COP_{HP,rev} & \text{reversible heat pump} \\ > COP_{HP,rev} & \text{impossible heat pump} \end{cases}$$